FIBRE PRODUCTS IN THE CATEGORY OF SCHEMES OVER S WITH THE PH TOPOLOGY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied verbatim from the pinned Stacks Project chapter file topologies.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Span recomputation by label pairing (LANE_BRIEF 10.2): the official pool candidate recorded statement [2884, 2892] and proof(s) [[2894, 2907]]; this spec covers the complete environment instead (statement 2884-2892, proof span(s) [(2894, 2907)]), since in this chapter the author's proof environment(s) follow the statement environment and some pool entries under-counted them.

Source reference. Original reference 000R: sources/stacks-chow/tags.txt, source lines 36–36. Cross-chapter standard prerequisite cited by the author inside the counted span; it is used as a pointer and its own proof is not part of this item.

Lemma 0.1. *Let S be a scheme. Let Sch_{ph} be a big ph site containing S. The underlying categories of the sites Sch_{ph}, $(Sch/S)_{ph}$, and $(Aff/S)_{ph}$ have fibre products. In each case the obvious functor into the category Sch of all schemes commutes with taking fibre products. The category $(Sch/S)_{ph}$ has a final object, namely S/S.*

Proof. For Sch_{ph} it is true by construction, see Sets, Lemma 000R. Suppose we have $U \to S$, $V \to U$, $W \to U$ morphisms of schemes with $U, V, W \in \mathrm{Ob}(Sch_{ph})$. The fibre product $V \times_U W$ in Sch_{ph} is a fibre product in Sch and is the fibre product of V/S with W/S over U/S in the category of all schemes over S, and hence also a fibre product in $(Sch/S)_{ph}$. This proves the result for $(Sch/S)_{ph}$. If U, V, W are affine, so is $V \times_U W$ and hence the result for $(Aff/S)_{ph}$. □

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